

Exercise 1) You can now enter some code for python to run:

```
>>> print("Hello world")
```

What does the print() do?. What's it in python?

```
>>> print("Hello World")
Hello World
```

It's the way to write characters (string) on some writable place, default is stdout, but it can write to file as well.

Exercise 2) Now, try to enter the following code in console to print Hello World and Your name.

```
>>> print(
... "Test123"
... );
```

What is happened and why?. Let's try another example below:

```
>>> print("Hello world"); \
... print("My name is ");
```

What does the \ do?.

```
>>> print("Hello World"); \
... print("My name is Phuriwat");
Hello World
My name is Phuriwat
```

Normally, Python in REPL mode execute line by line. “; \” make shell understand that user want to execute multiple line at once

Exercise 3) Strings can be combined together as follow.

```
>>> print("Hello world" + "This is python class")
```

Python does not add any whitespace between strings. What should we do?

There are multiple ways to do this. Assuming both input string is constant.

```
>>> print("Hello World" + " " + "This is python class")
Hello World This is python class
```

1. Manually add white space between both strings.

```
>>> print("Hello World", "This is python class")
Hello World This is python class
```

2. Use comma to pass tuple to print function.

```
>>> print("Hello World", end=" "); print("This is python class")
Hello World This is python class
```

3. Change "end" argument of first print call to "", then print second string

Exercise 4) Next, try the following command.

```
>>> print(len("Hello world" + " " + "This is python class")); \
... print(len("Helloworld"))
```

What are the outputs and what does the len() do?

```
>>> print(len("Hello world" + " " + "This is python class")); \
... print(len("Helloworld"))
32
10
```

len function receive iterator and return size of that iterator. In this case string is iterator of character and size of string is number of characters.

Exercise 5) Now, write the code to calculate summation of lengths of two String outputs from Exercise 4 together. What is the result?

```
>>> print(len("Hello world" + " " + "This is python class") + len("Helloworld"));
42
```

Exercise 6) In python, variable can be initialized and assigned the value by:

```
>>> fruit = "Mango";
>>> print(fruit);
```

Once the variable is assigned the value, it can be updated by setting it to a new value

```
>>> fruit = "Orange";
```

What is the output?

```
>>> fruit = "Mango";  
>>> print(fruit);  
Mango  
>>> fruit = "Orange";  
>>> print(fruit)  
Orange
```

After second assignment, value of fruit change to "Orange".

Exercise 7) Can we do the following commands? Why or Why not?

```
>>> print(animal);  
>>> animal = "Cat";
```

```
>>> print(animal);  
Traceback (most recent call last):  
  File "<stdin>", line 1, in <module>  
NameError: name 'animal' is not defined  
>>> animal = "Cat";
```

Variable cannot be reference before assignment.

Exercise 8) Try printing out the following numbers

```
>>> print(5);\n>>> print(1.5);\n>>> print(True);
```

```
>>> print(5);\n... print(1.5);\n... print(True);  
5  
1.5  
True
```

Exercise 9) Try the following code.

```
>>> x = "100";\  
>>> y = "250";\  
>>> print(x+y);
```

What is the result? Is it correct? If not make it correct.

```
>>> x = "100";\
... y = "250";\
... print(x + y);
100250
```

It is correct in term of concatenation of string, but not for numerical addition. For numerical addition both variable x and y must be number type (it is not specific which one) before adding.

```
>>> x = 100;\
... y = 250.;\
... print(x + y);
350.0
```

Exercise 10) Next, Let's try the following command.

```
>>> year = 2018 ; faculty_name = 'ICT';
>>> print("The faculty of {faculty_name} year {year}");
>>> print(f"The faculty of {faculty_name} year {year}");
```

What are the outputs and what does the f do?

```
>>> year = 2018; faculty_name = "ICT";\
... print("The faculty of {faculty_name} year {year}");\
... print(f"The faculty of {faculty_name} year {year}");
The faculty of {faculty_name} year {year}
The faculty of ICT year 2018
```

print(f...) function is like printf in C programming. Formatting variable into existing string.

Exercise 11) Determine the output of the following code:

```
>>> x = 10.25
>>> y = "15"
>>> print(int(x), int(y))
>>> y = int(x)
>>> x = int(y)
>>> print(x, y)
>>> x = 5
>>> print(x, y)
```

```

>>> x = 10.25
>>> y = "15"
>>> print(int(x), int(y))
10 15
>>> y = int(x)
>>> y = int(y)
>>> print(x, y)
10.25 10
>>> x = 5
>>> print(x, y)
5 10

```

Exercise 12) At the Python shell, you can now enter some code for python to run:

```

>>> import os
>>> os.system('mkdir c:\\cs159');
>>> file = open("c:\\cs159\\testfile.txt", "w");
>>> file.write("Hello, This is my file");
>>> file.write("\n--- newline ---");
>>> file.close();

```

What are the outputs and what does the open and write command do? What is the number after typing file.write command mean?

```

>>> import os
>>> os.system("mkdir c:\\cs159")
0
>>> file = open("C:\\cs159\\testfile.txt", "w")
>>> file.write("Hello, This is my file")
22
>>> file.write("\n--newline--")
13
>>> file.close()

```

file.write is writing a argument string to file object. Method write return number of character written into self.

Exercise 13) Next, try another following command

```

>>> import os
>>> file = open("c:\\cs159\\testfile.txt", "r");
>>> read_1 = file.read();
>>> read_2 = file.read(5);
>>> read_3 = file.readline();

```

```
>>> read_4 = file.readline();
>>> file.close();
```

Then, use print() command to display all variable named read 1, read 2, read 3, and read 4. What are the output? How each read command are different?

```
>>> import os
>>> file = open("C:\\cs159\\testfile.txt", "r")
>>> read_1 = file.read()
>>> read_2 = file.read(5)
>>> read_3 = file.readline()
>>> read_4 = file.readline()
>>> file.close()
>>> print(read_1)
Hello, This is my file
--nextline--
>>> print(read_2)

>>> print(read_3)

>>> print(read_4)
```

Seem, when open file there is pointer that point to the first character of the file, when file.read it will read the whole file till pointer found "EOF" then it will stop (I won't reset the pointer).

I believe file.read(5) is read only first 5 characters after pointer. file.readline() is read till it found newline (\n). Both commands will move pointer as well.

Exercise 14) Next, try another following command

```
>>> file = open("c:\\cs159\\testfile.txt", "a");
>>> file.write("\n -- append --");
>>> file.close();
```

Then, use print() command to display the file information. What are the different between open file with 'w', 'r', and 'a' parameter?

```
>>> file = open("C:\\cs159\\testfile.txt", "a")
>>> file.write("\n -- append --")
14
>>> file.close()
>>> file = open("C:\\cs159\\testfile.txt", "r")
>>> print(file.read())
Hello, This is my file
--nextline--
-- append --
```

Seem, mode "a" is append. When I open file using "a" mode, it will point pointer to the end of the file, when write is executed it will write string after the contain of the file.

Exercise 15) Write down the following code in Python editor. Then save as 'firstProg.py' on 'C:\\cs159' (Note that: at the bottom of the save as page, don't forget to select all file instead of txt file).

```
print("This is my first Python program");
num1 = 2;
num2 = 3;
sum = num1 + num2;
print(f"The result of {num1} + {num2} = {sum}");
```

then run it on script mode.

```
C:\cs159 python c:\cs159\firstProg.py
This is my first Python program
The result of 2 + 3 = 5
```

Exercise 16) Create a new Python script file named 'lab01.py'. Write a program to convert Celsius To Fahrenheit and Fahrenheit to Celsius. Ask user to 'Enter temperature in Celsius : ' then convert to Fahrenheit. Then verify your code by convert Fahrenheit back to Celsius:-
Celsius to Fahrenheit $F = \frac{9}{5} (C) + 32$

Test input 33.5 in Celsius is 92.3 Fahrenheit.

```
C:\cs159 python .\lab01.py
Enter tempurature in Celsius
33.5
92.30000000000001
```